

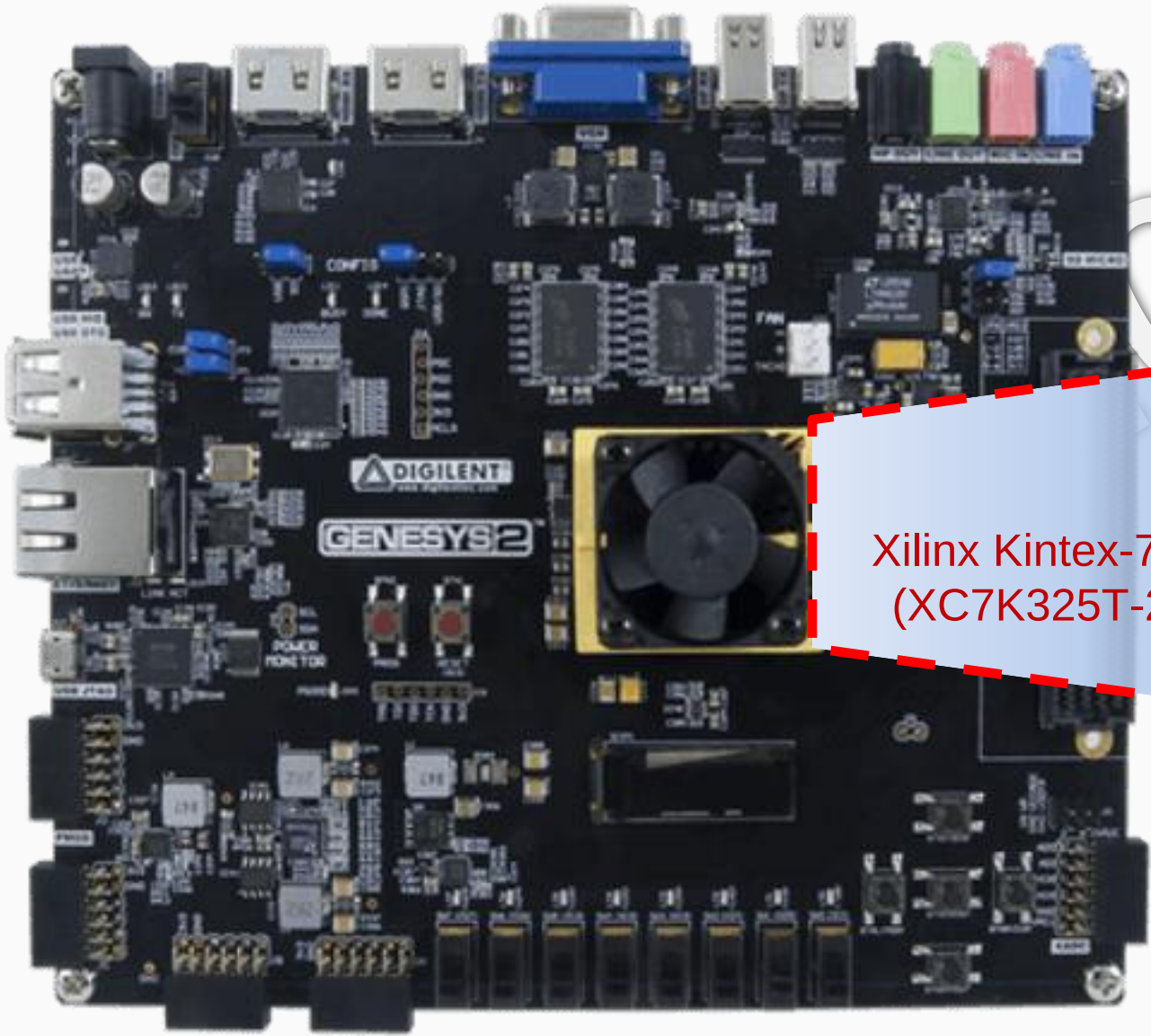
VEGA Microprocessor (RISC-V) based Design & Verification Challenge

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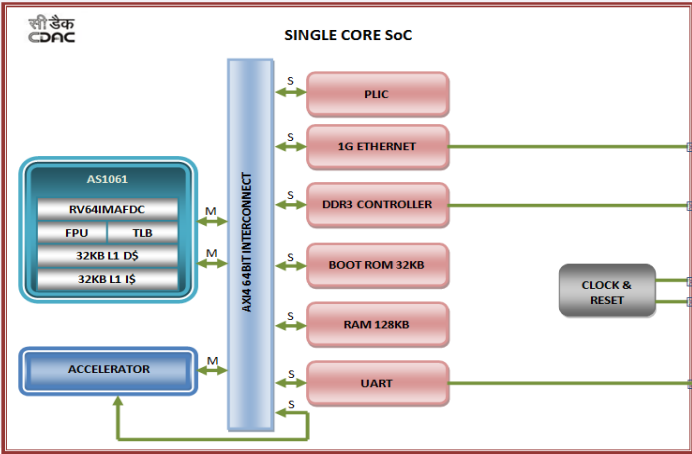
As part of the DIR-V Program and funding from Ministry of Electronics and Information technology (MeitY), The **Centre for Development of Advanced Computing (C-DAC)** developed a series of Microprocessors under the name “VEGA Processors”.

C-DAC has successfully developed VEGA series of microprocessors and related IPs and the complete ecosystem to enable fully indigenous product development that meets various requirements in the strategic, industrial and commercial sectors

These high performance processors are based on the open source RISC-V Instruction Set Architecture

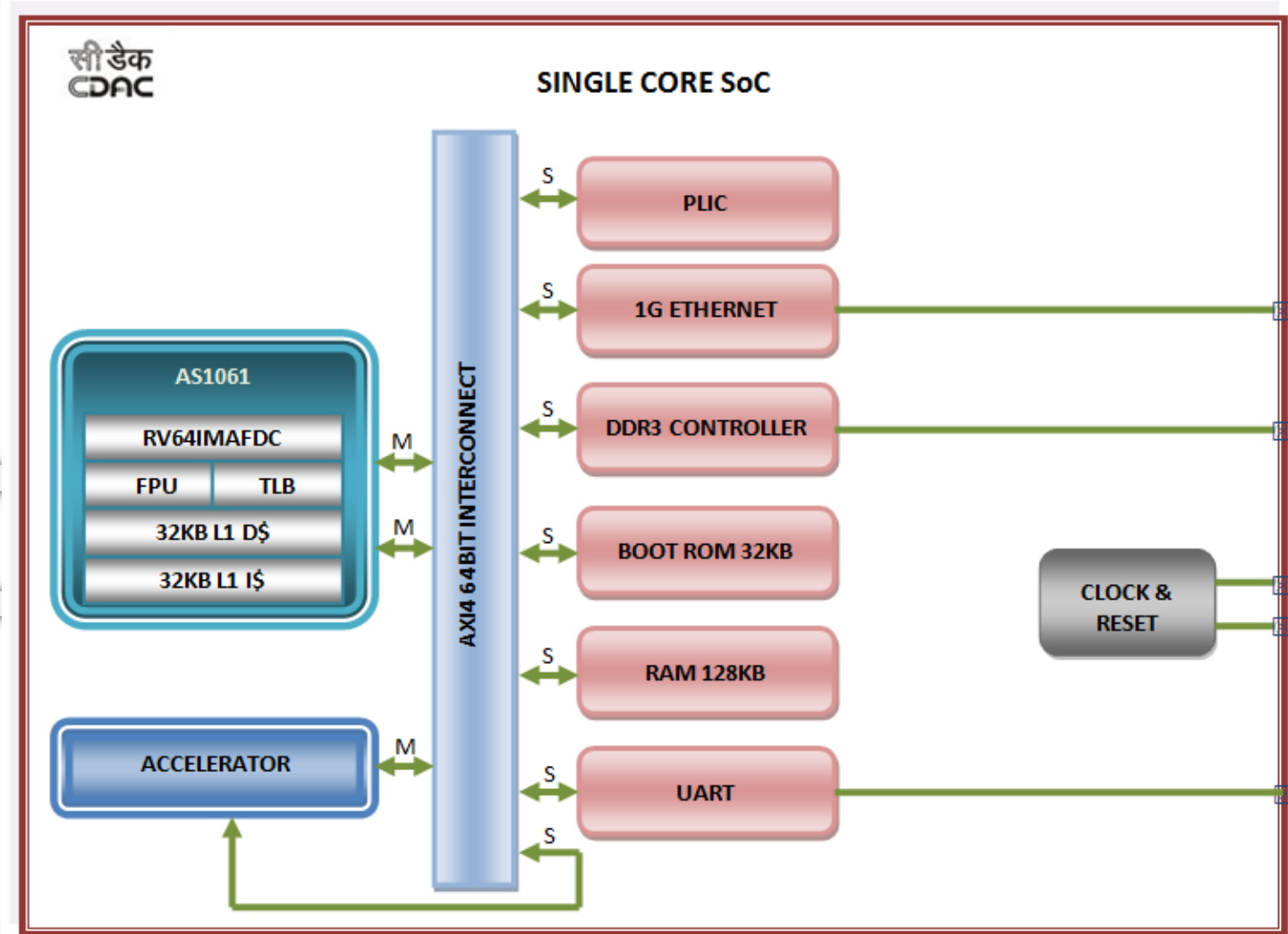


Xilinx Kintex-7 FPGA
(XC7K325T-2FFG)



Key features

- VEGA AS1061 Core
- 1GB DDR3 RAM
- 1 Gigabit Ethernet
- PLIC
- 32 KB ROM
- 128 KB RAM
- UART
- **Accelerator**



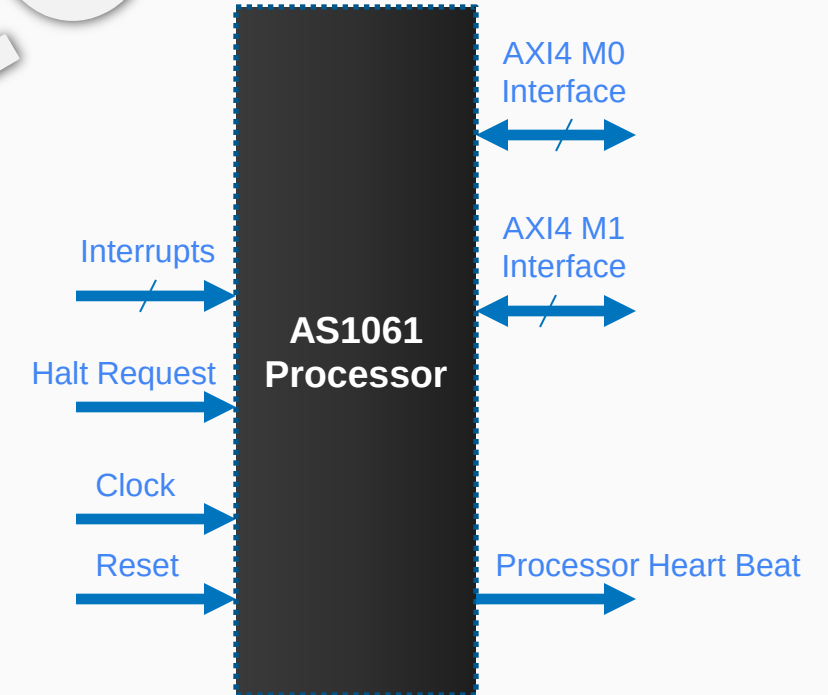
AS1061 Processor

Key Features :

- AXI4 Interface
- RISC-V 64GC (RV64IMAFDC) Instruction Set Architecture
- 6 stage in-order pipeline implementation
- 16/32-bit mix instruction format for better code density
- Memory Management Unit (MMU) and Physical Memory Protection (PMP)
- Branch predictor: BTB, BHT, RAS
- 32 KB L1 ICACHE & DCACHE
- Supports OpenOCD based debug

Core Configuration:

- CLK Frequency : 50 MHz



Key Features :

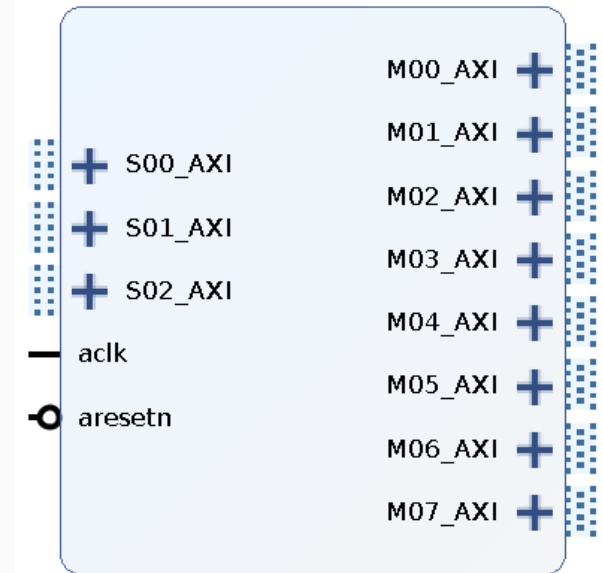
- AXI4 Interface

Core Configuration:

- CLK Frequency : 50 MHz
- Number of Slave & Master Interfaces : 3 & 8
- Protocol: AXI4
- Address Width, Data Width & ID width : 64, 64 & 12
- Master Interface Address : Give the Base Address and corresponding address width

For Reference:

- 👉 AXI Interconnect v2.1 LogiCORE IP Product Guide
- 👉 <https://users.ece.utexas.edu/~mcdermot/arch/articles/Zynq/pg059-axi-interconnect.pdf>



Key Features :

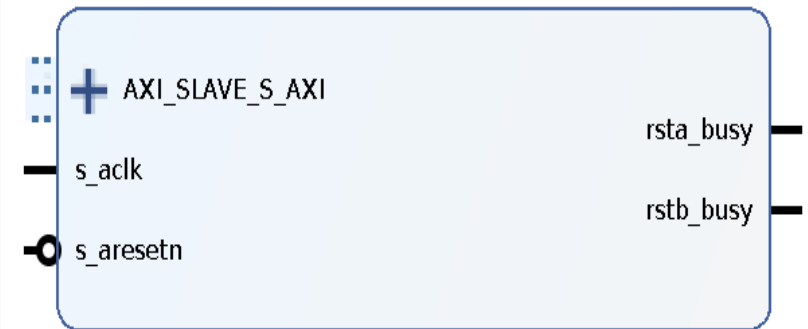
- AXI4 Interface
- Native interface core generates Single-Port RAM, Simple Dual-Port RAM, True Dual-Port RAM, Single-Port ROM, or Dual-Port ROM

Core Configuration:

- CLK Frequency : 50 MHz
- Memory Type: Simple Dual-Port RAM
- Algorithm & Primitives : Fixed Primitives & 2kx9
- AXI Type, AXI Slave Type & ID width : AXI4, Memory slave & 12
- Memory Size : Port A Width -64, Port A Depth -4096

For Reference:

- 👉 Block Memory Generator v8.4 LogiCORE IP Product Guide
- 👉 <https://docs.amd.com/v/u/en-US/pg058-blk-mem-gen>



Key Features :

- AXI4 Interface
- Native interface core generates Single-Port RAM, Simple Dual-Port RAM, True Dual-Port RAM, Single-Port ROM, or Dual-Port ROM

Core Configuration:

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- Memory Size : Port A Width -64, Port A Depth -16384

For Reference:

- 👉 Block Memory Generator v8.4 LogiCORE IP Product Guide
- 👉 <https://docs.amd.com/v/u/en-US/pg058-blk-mem-gen>



Key Features :

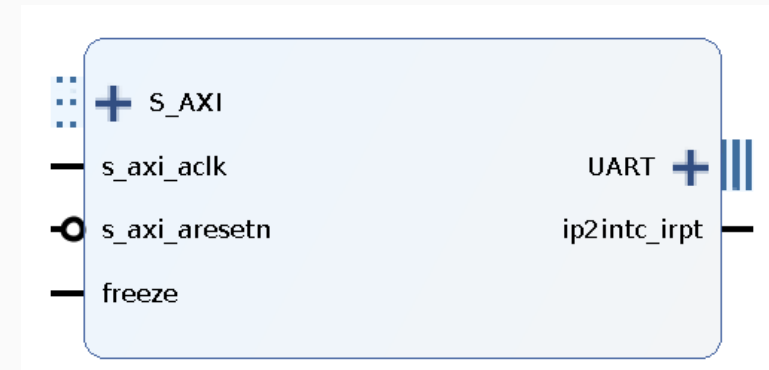
- AXI4-Lite interface for register access and data transfers
- Hardware and software register compatible with all standard 16450 and 16550 UARTs

Core Configuration:

- UART Mode : 16450
- CLK Frequency : 50 MHz

For Reference:

- 👉 AXI UART 16550 v2.0 LogiCORE IP Product Guide
- 👉 <https://docs.amd.com/v/u/en-US/pg143-axi-uart16550>

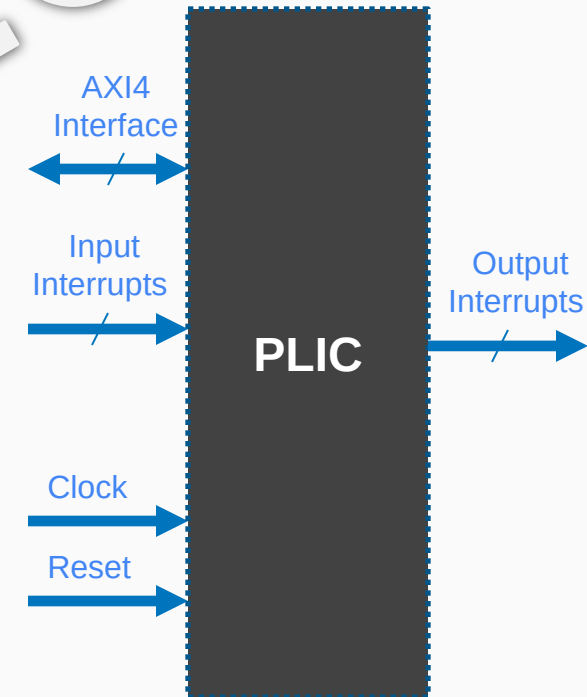


Key Features :

- AXI4 Interface
- Enable/ Disable property for individual interrupt sources per target
- Configurable support for connection up to 127 external interrupts.
- Supports both edge and level (E/L) triggered interrupt signals.
- Supports 15 priority levels for each interrupt source.

Core Configuration:

- CLK Frequency : 50 MHz
- Address Width, Data Width, Burst Size & Burst Length: 64, 64, Fixed & 0

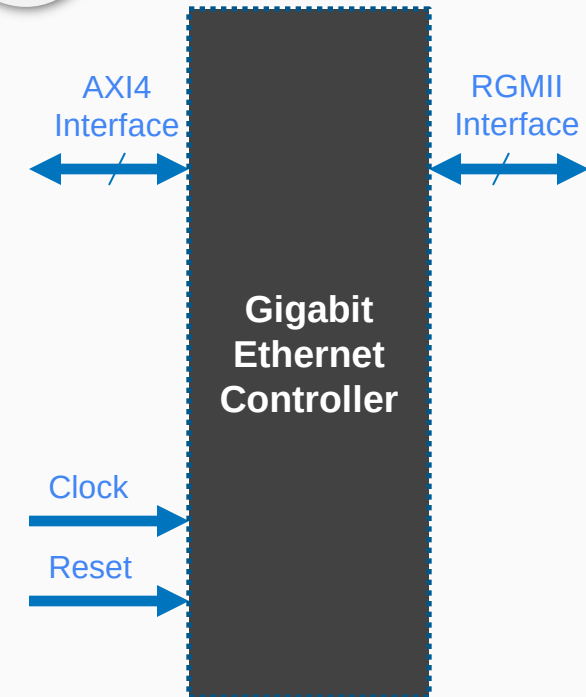


Key Features :

- AXI4 Interface
- Compliance to IEEE std. 802.3-2005
- RGMII interface to PHY
- Supports IEEE 802.3x pause frame based flow control for full duplex
- MDIO interface to manage PHY (MII Management)
- Supports various address filtering modes

Core Configuration:

- System CLK Frequency : 50 MHz
- GMAC CLK Frequency : 125 MHz



Key Features :

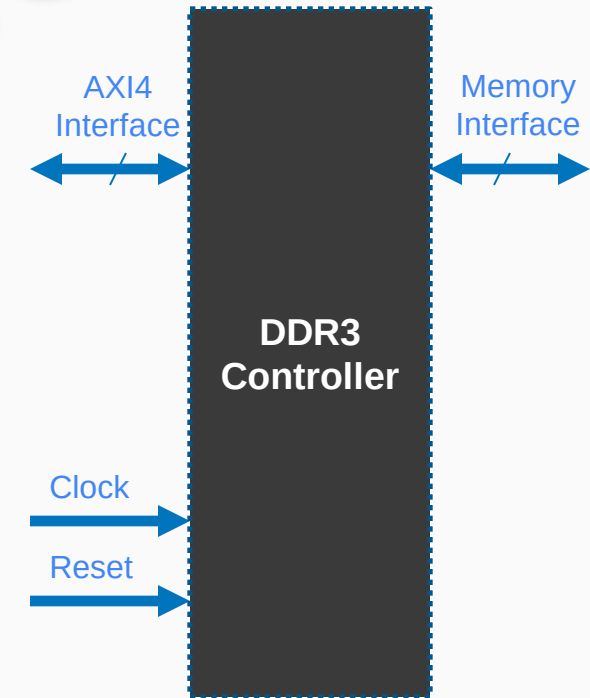
- AXI4 Interface

Core Configuration:

- System CLK Frequency : 50 MHz
- AXI4 Interface: Enable the AXI4 Interface
- Controller Type:DDR3 SDRAM
- Clock Period, Memory Part & Data Width : 1250ps, MT41J256m16XX-107 & 32
- Input Clock Period & Memory Address Mapping Selection : 5000ps(200MHz) & Bank | Row | Column

For Reference:

- 👉 7 Series FPGAs Memory Interface Solutions User Guide
- 👉 <http://www.bdtic.com/download/Xilinx/UG586.pdf>

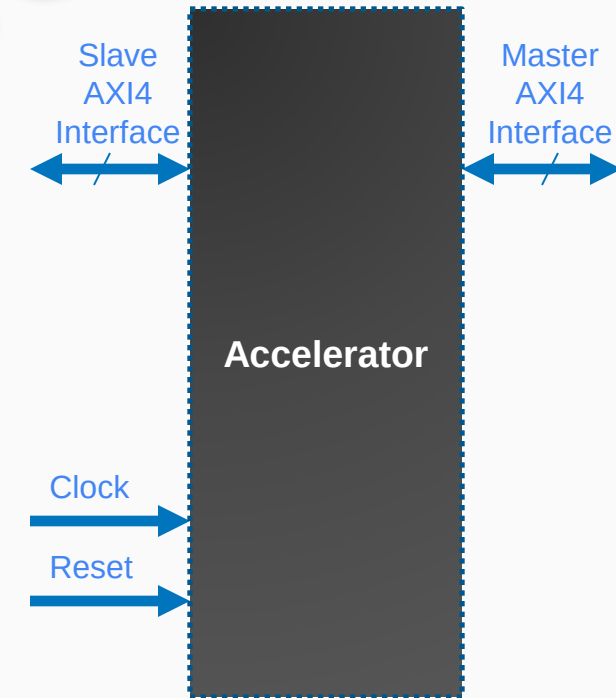


Key Features :

- AXI4 Interface

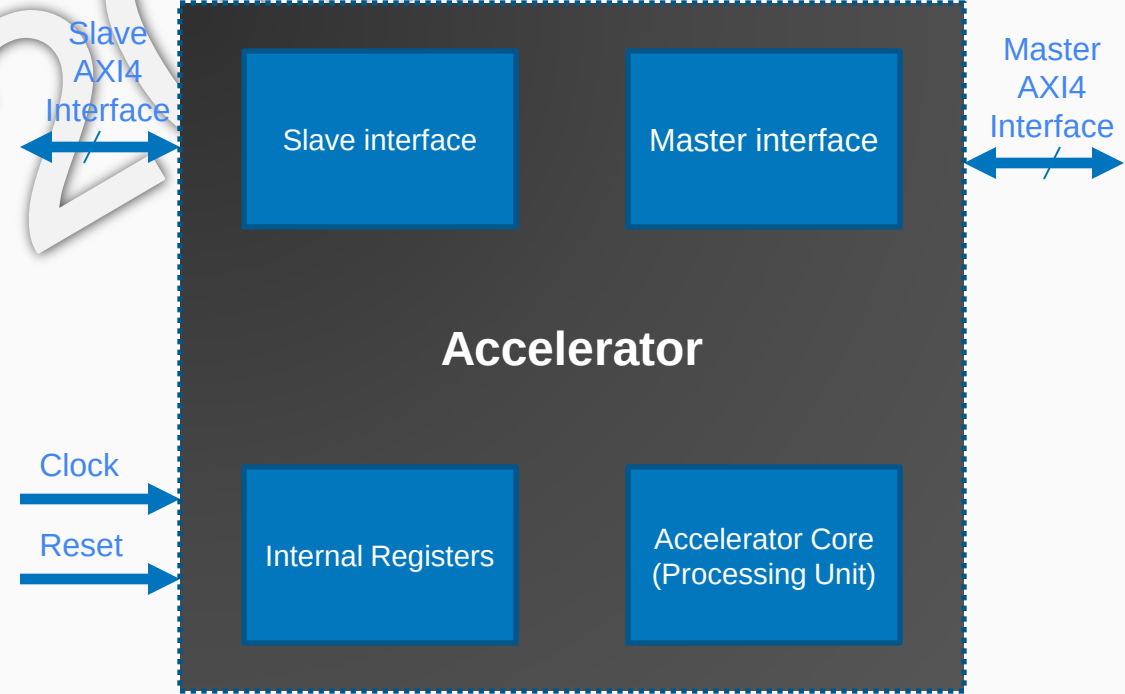
Core Configuration:

- CLK Frequency : 50 MHz
- Number of Slave & Master Interfaces : 1 & 1
- Address Width & Data Width : 64 & 64
- ID Width : 12
- Burst Type : Fixed, INCR, WRAP
- Burst Size : 8 Bytes
- Burst Length : 8 Bits



Register Details :

Offset	Functionality
00	Start/Ready
04	Reserved
08	Reserved
0c	Reserved
10	Matrix A Address- LSB
14	Matrix A Address- MSB
18	Reserved
1c	Matrix B Address- LSB
20	Matrix B Address- MSB
24	Reserved
28	Matrix C Address- LSB
2c	Matrix C Address- MSB
30	Reserved



How to connect and test a AXI4 accelerator to a VEGA SoC ?

1. Design the AXI4 Accelerator:

- Design a AXI4-compliant accelerator IP core that meets your application requirements.
- Data width (64 Bits), address width (64 Bits), ID width (12 Bits), burst type (Fixed, INCR, WRAP), burst size (8 Bytes), burst length (8 Bits), and other settings according to the specifications of the VEGA SoC.
- Resource utilization must be within the limits

Current resource utilisation

- Slice LUTs 55%
- Registers 20%
- Block RAM 15%
- DSP 10%

How to connect and test a AXI4 accelerator to a VEGA SoC ?

2. Understanding AXI4 Interfaces:

- The AXI4 (Advanced eXtensible Interface 4) protocol, part of the ARM AMBA specification, is widely used for high-performance, high-frequency system-on-chip (SoC) interconnects.
- In the AXI4 protocol, communication takes place between a Master (typically a processor, DMA controller, or accelerator) and a Slave (typically memory, peripheral, or IP block). The Master initiates all transactions, while the Slave responds.
- Channels in AXI4
 - ✓ Write Address (AW)
 - ✓ Write Data (W)
 - ✓ Write Response (B)
 - ✓ Read Address (AR)
 - ✓ Read Data (R)

Each channel operates independently with handshake (VALID / READY) for high throughput.

How to connect and test a AXI4 accelerator to a VEGA SoC ?

3. SoC Integration:

- **Address Mapping:**

- Allocate address space for the following peripherals using the VEGA SoC's defined memory map, ensuring no overlaps and proper alignment.

Vega SoC Address Map		
Component	AS1061: Single Core SoC	
	Start Address	End Address
Boot ROM	0x0001_0000	0x0001_7FFF
Internal RAM	0x0002_0000	0x0003_FFFF
DDR3 CONTROLLER	0x8000_0000	0xBFFF_FFFF
1G ETHERNET	0x2003_0000	0x2003_FFFF
ACCELERATOR	0x2004_0000	0x2004_FFFF
UART	0x1000_1000	0x1000_1FFF

- Define the base address and address range for the accelerator to support both memory accesses and register mappings. Memory accesses (e.g., data load/store to DDR) are performed via the AXI master interface, while register mappings (e.g., control and status registers) are accessed through the AXI slave interface.

How to connect and test a AXI4 accelerator to a VEGA SoC ?

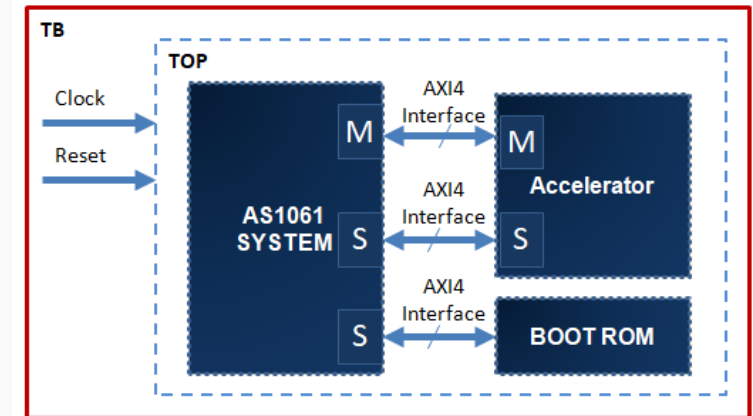
4. IP Core Integration

- **IP Integration:**

- Integrate the RTL code of the accelerator IP core into the VEGA SoC.
- Connect the accelerator's AXI4 slave interface to the master port of the AXI4 interconnect, and its AXI4 master interface to the slave port of the interconnect within the VEGA SoC. These dedicated interconnect ports for the accelerator can be located in the VEGA SoC Top module.

- **Clock and Reset Domain:**

- Ensure the accelerator IP core is properly synchronized with the VEGA SoC's 50 MHz clock and reset domain to maintain reliable integration.



How to connect and test a AXI4 accelerator to a VEGA SoC ?

VEGA SoC AXI4 Interface Signal

AXI4 Master Interface

Master Write Address Channel

m_axi_awvalid	1 bit
m_axi_awid	12 bits
m_axi_awlen	8 bits
m_axi_awsz	3 bits
m_axi_awburst	2 bits
m_axi_awlock	1 bit
m_axi_awcache	4 bits
m_axi_awqos	4 bits
m_axi_awaddr	64 bits
m_axi_awprot	3 bits
m_axi_awready	1 bit

Master Write Data Channel

m_axi_wvalid	1 bit
m_axi_wlast	1 bit
m_axi_wdata	64 bits
m_axi_wstrb	8 bits
m_axi_wready	1 bit

Master Write Response Channel

m_axi_bready	1 bit
m_axi_bvalid	1 bit
m_axi_bid	12 bits
m_axi_bresp	2 bits

Master Read Address Channel

m_axi_arvalid	1 bit
m_axi_arid	12 bits
m_axi_arlen	8 bits
m_axi_arsz	3 bits
m_axi_arburst	2 bits
m_axi_arlock	1 bit
m_axi_arcache	4 bits
m_axi_arqos	4 bits
m_axi_araddr	64 bits
m_axi_arprot	3 bits
m_axi_arready	1 bit

Master Read Data Channel

Signal	Width
m_axi_rready	1 bit
m_axi_rvalid	1 bit
m_axi_rid	12 bits
m_axi_rlast	1 bit
m_axi_rresp	2 bits
m_axi_rdata	64 bits

AXI4 Slave Interface

Slave Write Address Channel

s_axi_awid	12 bits
s_axi_awaddr	64 bits
s_axi_awlen	8 bits
s_axi_awsz	3 bits
s_axi_awburst	2 bits
s_axi_awlock	1 bit
s_axi_awcache	4 bits
s_axi_awprot	3 bits
s_axi_awqos	4 bits
s_axi_awvalid	1 bit
s_axi_awready	1 bit

Slave Write Data Channel

s_axi_wdata	64 bits
s_axi_wstrb	8 bits
s_axi_wlast	1 bit
s_axi_wvalid	1 bit
s_axi_wready	1 bit

Slave Write Response Channel

s_axi_bready	1 bit
s_axi_bid	12 bits
s_axi_bresp	2 bits
s_axi_bvalid	1 bit

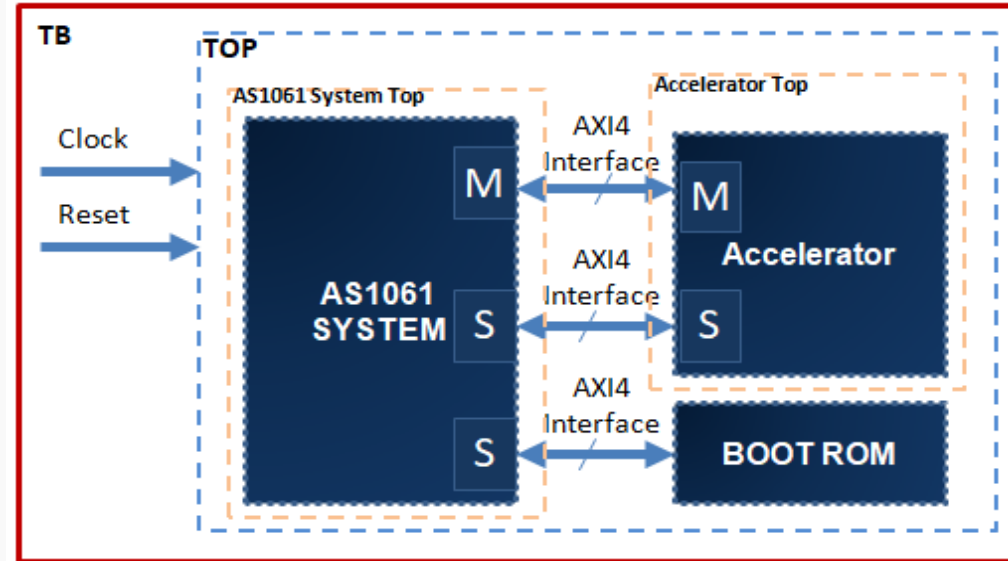
Slave Read Address Channel

s_axi_arid	12 bits
s_axi_araddr	64 bits
s_axi_arlen	8 bits
s_axi_arsz	3 bits
s_axi_arburst	2 bits
s_axi_arlock	1 bit
s_axi_arcache	4 bits
s_axi_arprot	3 bits
s_axi_arqos	4 bits
s_axi_arvalid	1 bit
s_axi_arready	1 bit

Slave Read Data Channel

s_axi_rready	1 bit
s_axi_rid	12 bits
s_axi_rdata	64 bits
s_axi_rresp	2 bits
s_axi_rlast	1 bit
s_axi_rvalid	1 bit

How to connect and test a AXI4 accelerator to a VEGA SoC ?



Using the same data width and interface type allows direct connectivity between the accelerator and the VEGA SoC via the AXI4 interface.

- Use simulation tools to verify the integrated accelerator's functionality with the VEGA SoC.
- Validate data transfers, register access, and error-handling scenarios to ensure correct operation.

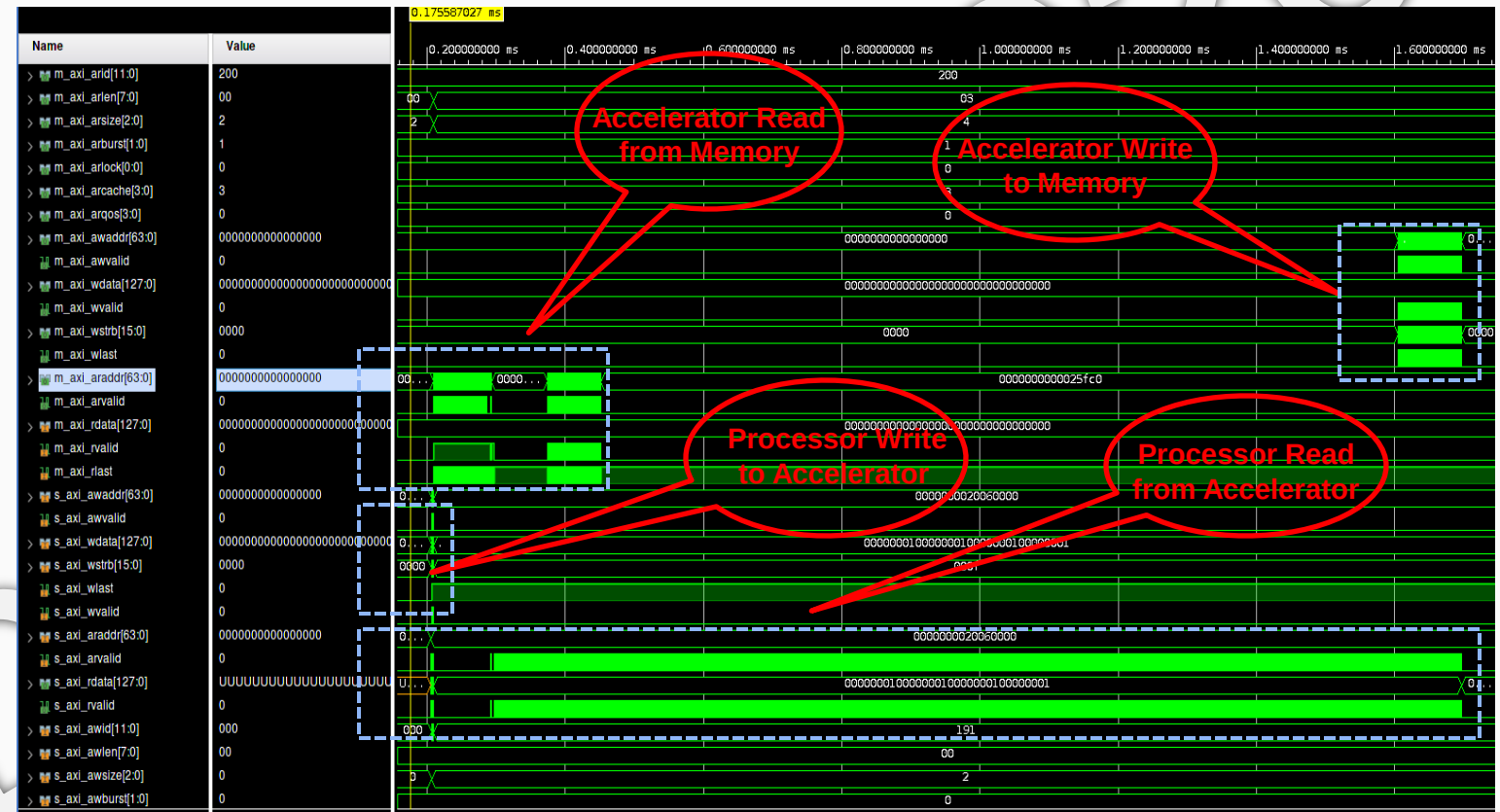


Figure: Data transfers to and from memory are carried out via the AXI Master interface, whereas register-level communication with the accelerator is performed through the AXI Slave interface.



ANY QUESTION?
/ THANK YOU /

ADVANCED
PROCESSING UNIT

4-CORE CPU /
2.0GHz

VEGA

DDRS

32GB

DR5

DDRS